Cognitive Thermodynamics, The Ariṣaḍvarga, and the Landauer Veto

> **Cognitive & Psychological Focus:** This treatise formalizes **Tier III Cognitive Architectures**, **Bayesian Active Inference Engines**, the **Thermodynamics of the Six Internal Afflictions (*Ariṣaḍvarga*)**, and the **Landauer Information-Erasure Cost of Voluntary Veto (*Viveka*)**.

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1. Tier III: Cognitive Agents as Complex Predictive Engines

In draft.md §2.3.4 & §4, a cognitive agent ($E \in T_{\text{III}}$) operates across complexified phase space $\Omega_{\mathbb{C}} = \Omega_{\mathbb{R}} \oplus i\Omega_{\mathfrak{Im}}$: * **Real Substrate ($\Omega_{\mathbb{R}}$):** Immediate somatic, motor, and neurochemical configurations. * **Imaginary Substrate ($\Omega_{\mathfrak{Im}}$):** Internal generative Bayesian models simulating counterfactual future trajectories ($\hat{\mathbf{C}}_{\text{future}}$).

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                          THE COGNITIVE ENERGY ALLOCATION FORK
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                          Total Informational Fuel: Ė_total
                                           │
             ┌─────────────────────────────┴─────────────────────────────┐
             │                                                           │
[REAL MOTOR ACTUATION: Ė_Re]                                [IMAGINARY SIMULATION: Ė_Im]
• Immediate somatic response                                • Counterfactual future forward simulation
• Physical muscular work                                    • Sagawa-Ueda negentropy bound:
• χ* ≈ 1: Optimal balance                                   • ⟨W_dissipated⟩ ≥ Δℱ_noneq - k_B T Δℐ(Ĉ; C)
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The Sagawa-Ueda Information-Thermodynamic Bound:

The dissipation required to survive an environmental shock is reduced by the mutual information $\Delta\mathcal{I}(\hat{\mathbf{C}}; \mathbf{C}_{\text{future}})$ acquired via internal simulation:

$$\boxed{\langle W_{\text{dissipated}} \rangle \geq \Delta\mathcal{F}_{\text{noneq}} - k_B T \cdot \Delta\mathcal{I}\left(\hat{\mathbf{C}}_{\text{predicted}}; \mathbf{C}_{\text{future}}\right)}$$

Pre-stiffening cognitive priors ($\mathbf{R}_{\text{active}}(t - \Delta t)$) reduces physical damage to near zero upon predictable impacts.

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2. Involuntary Latency (*Vāsanā*) vs. Voluntary Landauer Veto Gating (*Viveka*)

Living cognitive agents possess two distinct modes of interfacial actuation:

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                         THE ACTUATION PATHWAYS
                                   │
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         │                                                   │
[INVOLUNTARY REFLEX: VĀSANĀ]              [VOLUNTARY VETO: VIVEKA]
• Fast feed-forward reactive arc          • Slow Dyson time-ordered loop
• Response latency: τ ~ 10^(-2) s         • Pays Landauer erasure cost:
• Computation entropy: Ḣ ≈ 0             • Ė_veto = k_B T ln 2 · Ḣ_erasure > 0
• Automated habit execution               • Active inhibitory gating 𝒪_inhibit
```

- **Involuntary Reflex (*Vāsanā / Saṃskāra*):**
- An automated feed-forward reflex arc where incoming challenge traction $\mathbf{C}$ directly triggers motor expression $\boldsymbol{\mathcal{X}}$ through pre-configured neural pathways ($\tau_{\text{latency}} \sim 10^{-2}\,\text{s}$). Because no deliberative computation or state re-evaluation occurs, the Landauer informational entropy cost is zero ($\dot{\mathcal{H}} \approx 0$).

- **Voluntary Discriminative Veto (*Viveka*):**
- The conscious intervention of the higher-order Dyson loop ($\mathcal{T}\,\exp(\int \hat{\mathcal{L}} d\tau)$), which inspects the automated trajectory and applies an active inhibitory gate ($\mathcal{O}_{\text{inhibit}}$) to prevent expression.
- **The Landauer Metabolic Tax:** Erasing the automated motor command requires physical bit erasure, demanding continuous free-energy dissipation:

$$\boxed{\dot{\mathcal{E}}_{\text{veto}} = k_B T \ln 2 \cdot \dot{\mathcal{H}}_{\text{erasure}} > 0}$$

*Exercising conscious restraint (*Viveka / Dama*) is physically exhausting because it consumes real biochemical glucose ($k_B T \ln 2$) to clear neural register buffers.

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3. Top-Down Craving (*Kāma*) vs. Bottom-Up Hunger (*Kṣudhā*)

***Hunger (*Kṣudhā*):** A bottom-up First-Law exergy deficit occurring entirely in real configuration space $\Omega_{\mathbb{R}}$:

$$\dot{E}_{\text{fuel}} < \dot{E}_{\text{crit}} \equiv T_{\text{ambient}} \int_E \sigma_{\text{total}}\, dV \implies \frac{d\mathcal{G}}{dt} < 0$$

Kṣudhā is a biochemical indicator of metabolic depletion (falling blood glucose, low ATP/AMP ratio) requiring zero memory simulation.

***Desire / Craving (*Kāma / Tṛṣṇā*):** A top-down simulation executed in imaginary phase space $\Omega_{\mathfrak{Im}}$:

$$D_{\mathfrak{Im}} = \hat{\pi}(\mathcal{F}_{\text{ledger}}) \xrightarrow{\text{simulation}} \Delta\hat{\mathcal{G}}_{\text{predicted}} > 0$$

Kāma occurs when the cognitive model recalls past reward states from $\mathcal{F}_{\text{ledger}}$ and projects an artificial free-energy surplus. This top-down prediction transduces downward into physical physiology via the Semantic Transduction Tensor (

$\mathbf{C}_{\text{craving}} = \mathbf{K}_{\text{trans}} \cdot \nabla_\theta \mathcal{D}_{\text{KL}}$), generating visceral dopamine cascades and restlessness **even when fully satiated.**

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4. Thermodynamics of the *Ariṣaḍvarga* (The Six Internal Afflictions)

In classical Sanskrit psychology, the *Ariṣaḍvarga* represents the six inner enemies that destroy discernment. In our framework, these are derived as **exact mathematical distortions of the structural margin tensor** $\phi(x, t)$:

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                        THE ARIṢAḌVARGA TENSOR DISTORTIONS
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      ┌─────────────────────────────────┼─────────────────────────────────┐
      │                                 │                                 │
[KĀMA: PHANTASMAGORIC OVERDRIVE]  [KRODHA: OVER-REACTIVE STRESS]    [LOBHA: RESOURCE STRANGULATION]
Ė_Im ⟶ ∞, Ė_Re ⟶ 0                  ||R|| >> ||C|| (Self-Fracture)   J_metabolic ⟶ 0 (Local Gluttony)
      │                                 │                                 │
[MOHA: INVERTED PRIORS]           [MADA: HUBRISTIC BLINDNESS]       [MĀTSARYA: MALICIOUS SABOTAGE]
||Ĉ - C|| ⟶ ∞ (Boundary Delusion) Ĉ ≡ 0 (Brittle Impact Failure)   Consuming fuel to drop Δφ_AB
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- **Kāma (काम — Unbounded Imaginary Projection Overdrive):**
- Informational fuel allocation diverges ($\chi \to \infty \implies \dot{\mathcal{E}}_{\mathfrak{Im}} \to \dot{\mathcal{E}}_{\text{total}}, \dot{\mathcal{E}}_{\mathfrak{Re}} \to 0$). The entity expends all free energy simulating ungrounded imaginary prospects while the physical substrate starves into anergic boundary collapse.
- **Krodha (क्रोध — Explosive Over-Reactive Stress Discharge):**
- Internal counter-stress vastly exceeds external challenge ($\|\mathbf{R}_{\text{active}}\| \gg \|\mathbf{C}\|$). The localized stress concentration generates internal Rankine-Hugoniot shock waves that fracture the entity's own boundary from within.
- **Lobha (लोभ — Parasitic Resource Hoarding):**
- A constituent node hoards fuel measure ($\mu(E^j) \uparrow$) while choking metabolic syncytial flux ($\mathbf{J}^{j \to k}_{\text{metabolic}} \to 0$). This starves neighboring nodes, triggering a localized ischemia cascade that collapses the macro-envelope.
- **Moha (मोह — Perceptual Delusion / Model Inversion):**
- The generative model divergence explodes ($\|\hat{\mathbf{C}} - \mathbf{C}_{\text{true}}\| \to \infty$). The entity mistakes destructive threats for fuel sources, actively opening its boundary pores ($\mathbf{v}_n \cdot \hat{n} < 0$) to toxic foreign challenge fields.
- **Mada (मद — Hubristic Invulnerability / Zero-Challenge Prior):**
- The entity sets its predicted external challenge to zero ($\hat{\mathbf{C}} \equiv \mathbf{0}$), disarming its adaptive viscoelastic dampers ($\nu \to 0$). Upon the arrival of real environmental shocks, it undergoes catastrophic brittle fracture without damping.
- **Mātsarya (मात्सर्य — Malicious Zero-Sum Trophic Sabotage):**
- An entity expends its own metabolic exergy solely to diminish a neighbor's margin ($\Delta\phi_{AB} < 0$), lowering total syncytial free energy and precipitating mutual systemic lysis.